Unit 3

Question bank

Short Answer Questions

1. Explain how the Combs method helps in reducing the number of rules in a fuzzy inference system. Illustrate with an example for a 2-input fuzzy system, each with 3 fuzzy sets.
2. Use SVD to reduce a given fuzzy relation matrix and reconstruct it using top-2 singular values
3. How does fuzzy synthetic evaluation handle uncertainty in decision-making?
4. Why are weights important in fuzzy synthetic evaluation, and how are they determined?
5. What is fuzzy ordering and how does it differ from classical ordering?
6. How can fuzzy ordering handle situations with incomplete or imprecise preferences?
7. Give an example where nontransitive ranking might appear in real-world fuzzy decision problems.
8. What are the challenges of dealing with nontransitive ranking in fuzzy ordering?
9. How does the COMBS method improve decision-making in fuzzy environments?
10. Mention one application of SVD in fuzzy system modeling or evaluation.

11Explain the role of weighting in multi-objective decision-making.

12 How can fuzzy logic assist in achieving consensus among decision-makers?

Long answer questions

1. Define preference aggregation and consensus in the context of group decision-making. Explain how fuzzy logic can be used to model preferences and achieve consensus among decision-makers.
2. What is multi-objective decision making? Explain how fuzzy logic aids in solving multi-objective problems, especially when dealing with conflicting criteria.
3. Explain the concept of fuzzy ordering and how it improves decision-making over traditional crisp ordering. Discuss how nontransitive ranking arises in fuzzy ordering and its implications for decision analysis.

4. Describe fuzzy preference relations and explain how they can lead to nontransitive ranking. How can decision-makers address issues caused by nontransitivity in fuzzy systems
5. What is Singular Value Decomposition (SVD)? Explain its mathematical foundation and discuss its applications in data reduction and pattern recognition. How can SVD be useful in fuzzy system modeling?
6. Discuss how SVD can be applied in the context of multi-criteria decision-making or fuzzy evaluation methods. Include its advantages and potential limitations.
7. What is the COMBS method in the context of fuzzy decision-making? Describe its main components and the step-by-step procedure.
8. Explain how the COMBS method helps handle uncertainty and imprecision in multi-criteria decision-making. Provide an example of its practical application.
9. Define fuzzy ordering and explain its significance in multiobjective decision making. How does it help in ranking alternatives under conflicting objectives?